

Question:

The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

Answer Choices:

- A. follicle-stimulating hormone
- B. oxytocin
- C. pituitary gland
- D. hypothalamus
- E. gonadotropin-releasing hormone
- F. luteinizing hormone
- G. testosterone

Reasoning + Answer:

<think>
The feedback loop in men that regulates the production of sex hormones involves the interplay between the hypothalamus, pituitary gland, and the testes. The hypothalamus produces gonadotropin-releasing hormone (GnRH), which stimulates the pituitary gland to release follicle-stimulating hormone (FSH) and luteinizing hormone (LH). FSH and LH then stimulate the testes to produce sex hormones, including testosterone. The high levels of testosterone feed back to the pituitary gland and hypothalamus, inhibiting the release of GnRH, FSH, and LH. This creates a negative feedback loop that maintains the balance of sex hormone production. Oxytocin is primarily involved in social bonding and childbirth, rather than sex hormone production. Testosterone is a sex hormone produced by the testes, not a regulator of its production. Therefore, the correct answer is the one that is not directly involved in the feedback loop regulating sex hormone production in men. Considering the given options, oxytocin is the one that is not directly involved in the feedback loop. So, the reasoning process involves identifying the components of the feedback loop and understanding their roles, and then eliminating the options that are directly involved. The answer would be the one that is not involved in the feedback loop, which is oxytocin. Therefore, the correct answer would be B. oxytocin. </think><answer> B. oxytocin </answer>

